H

Weekly Progress Report

Sprint 4 — Explainability & Ethics (Steps 6–7)

SENG 430 · Software Quality Assurance · Cankaya University · Spring 2025–2026
Instructor: **Dr. Sevgi Koyuncu Tunç**

Group Name	HealthWithSevgi
Week Number	Week 8 (Final week of Sprint 4)
Scrum Master	Batuhan Bayazıt
Date Submitted	April 15, 2026
Sprint Period	March 31 – April 15, 2026
Domains Implemented	**20 / 20** (all clinical domains functional)

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GitHub: github.com/EudaLabs/HealthWithSevgi
Live Demo: 0xbatuhan4-healthwithsevgi.hf.space

1. Team

ROLE	NAME	STUDENT ID
Product Owner + Developer	Efe Çelik	202128016
UX Designer	Burak Aydoğmuş	202128028
Lead Developer + Scrum Master	Batuhan Bayazıt	202228008
Developer	Berat Mert Gökkaya	202228019
QA / Documentation Lead	Berfin Duru Alkan	202228005

2. Burndown Chart

> Sprint 4 duration: March 31 – April 15, 2026. Scope: Step 6 (Explainability) and Step 7 (Ethics & Bias). 26 total issues in sprint scope (10 User Stories from backlog + 16 new Sprint 4 stories). All 26 stories marked Done. 52 commits merged to main during the sprint period.

Screenshot from Jira Sprint 4 board showing remaining story points vs. ideal burndown line.

METRIC	VALUE
Sprint Duration	March 31 – April 15, 2026
Total Issues (Sprint 4 scope)	26 (10 backlog stories + 16 new stories)
Stories Completed	26 / 26
Done	26 issues — all Sprint 4 scope items
In Progress	0 issues
Commits	52 commits to main
Dev Completion	All Step 6 & Step 7 features delivered

3. Completed This Sprint

All features listed below are fully implemented and merged on the `main` branch. Completion evidence comes from Git commit history, Jira board, and UAT test results.

3.1 User Stories — Step 6 Explainability

STORY ID	TITLE	ASSIGNEE	COMPLETED
US-020 (SCRUM-45)	View feature importance chart with clinical names	Batuhan Bayazıt	2026-04-09
US-021 (SCRUM-46)	View SHAP waterfall explanation for individual patient	Berat Mert	2026-04-09
US-029 (SCRUM-142)	Display correlation vs. causation disclaimer on Step 6	Batuhan Bayazıt	2026-04-09

3.2 User Stories — Step 7 Ethics & Bias

STORY ID	TITLE	ASSIGNEE	COMPLETED
US-022 (SCRUM-47)	View subgroup fairness performance table with bias alerts	Berat Mert	2026-04-12
US-023 (SCRUM-48)	Complete EU AI Act compliance checklist	Berfin Duru Alkan	2026-04-12
US-024 (SCRUM-49)	View training data representation chart vs. hospital population	Efe Çelik	2026-04-12
US-025 (SCRUM-50)	Download PDF summary certificate after completing all steps	Efe Çelik	2026-04-13
US-027 (SCRUM-57)	View real-world AI failure case studies in Step 7	Efe Çelik	2026-04-13

3.3 User Stories — Sprint 4 Enhancements

STORY ID	TITLE	ASSIGNEE	COMPLETED
US-030 (SCRUM-199)	Interactive parallel coordinates chart for model comparison	Batuhan Bayazıt	2026-04-12
US-031 (SCRUM-200)	Sanitize inf/nan floats and optimize ML training pipeline	Berat Mert	2026-04-12
US-032 (SCRUM-201)	InsightService backend — LLM-powered clinical insights with MedGemma/Gemini	Berat Mert	2026-04-13
US-033 (SCRUM-202)	AI clinical assessment UI and enriched ethics modals	Efe Çelik	2026-04-14
US-040 (SCRUM-209)	UI/UX design for model comparison chart & clinical insights UI	Burak Aydoğmuş	2026-04-12

3.4 ISO 42001 Documentation Stories

STORY ID	TITLE	ASSIGNEE	COMPLETED
US-034 (SCRUM-203)	ISO 42001 Ch.1 — AIMS Scope, Context & Stakeholders	Efe Çelik	2026-04-14
US-035 (SCRUM-204)	ISO 42001 Ch.2 — AI Policy & Governance Roles	Burak Aydoğmuş	2026-04-14
US-036 (SCRUM-205)	ISO 42001 Ch.3 — Data Governance & Third-Party Management	Berat Mert	2026-04-14
US-037 (SCRUM-206)	ISO 42001 Appendix A (SoA) & Appendix B (Feedback Form)	Berfin Duru Alkan	2026-04-14

STORY ID	TITLE	ASSIGNEE	COMPLETED
US-038 (SCRUM-207)	ISO 42001 report technical review and cross-referencing	Batuhan Bayazıt	2026-04-14

3.5 QA / Test Stories

STORY ID	TITLE	ASSIGNEE	COMPLETED
TEST-S6 (SCRUM-65)	Test — Step 6 Explainability	Berfin Duru Alkan	2026-04-15
TEST-S7 (SCRUM-66)	Test — Step 7 Ethics & Bias + Certificate	Berfin Duru Alkan	2026-04-15
US-042 (SCRUM-211)	QA execution — Step 6 feature importance & clinical sense-check (TC-S4-001 to TC-S4-013)	Burak Aydoğmuş	2026-04-15
US-043 (SCRUM-212)	QA execution — Step 6 patient waterfall, what-if & domain coverage (TC-S4-014 to TC-S4-030)	Batuhan Bayazıt	2026-04-15
US-044 (SCRUM-213)	QA execution — Step 7 subgroup table & bias detection (TC-S4-031 to TC-S4-040)	Berat Mert	2026-04-15
US-045 (SCRUM-214)	QA execution — Step 7 EU AI Act checklist, representation & case studies (TC-S4-041 to TC-S4-054)	Efe Çelik	2026-04-13
US-046 (SCRUM-215)	QA execution — Certificate PDF validation & cross-step consistency (TC-S4-055 to TC-S4-069)	Berat Mert	2026-04-15
US-047 (SCRUM-216)	QA execution — End-to-end pipeline & deliverable verification (TC-S4-070 to TC-S4-084)	Berfin Duru Alkan	2026-04-15

Total Stories Delivered: 26

4. In Progress

None — all Sprint 4 stories are completed and marked as Done in Jira.

5. Blocked / At Risk

No blockers. All Sprint 4 deliverables are complete.
All 26 Jira stories have been transitioned to **DONE** status.

6. Key Decisions Made

1. **Clinical-first language throughout Steps 6–7.** All feature labels use plain clinical names (e.g., "Serum Creatinine (mg/dL)") instead of raw database column names. A 100+ entry `CLINICAL_NAME_MAP` translates every feature. Zero raw column names visible to the end user.

2. **Fast What-If analysis with predict_proba.** Probability shifts are calculated using `predict_proba` instead of recomputing SHAP values, making What-If simulations near-instant (<100ms) compared to 3–5 seconds for full SHAP recomputation.

3. **Three-patient sampling strategy for waterfall explanations.** Low-risk, mid-risk, and high-risk representative patients are sampled automatically from test data. This provides meaningful contrast in SHAP waterfall explanations without requiring the user to browse through hundreds of patients.

4. **Strict bias thresholding for fairness alerts.** The bias auto-detection banner uses a >10pp sensitivity gap rule for triggering warnings, and a >15pp demographic gap for representation warnings. These thresholds were chosen to balance clinical significance against false alarms.

5. **LLM-powered clinical insights via InsightService.** Integrated MedGemma and Gemini as LLM providers for generating domain-specific clinical interpretations of model results, enriching the EU AI Act checklist descriptions and case study analysis.

7. Test Results

Based on the Sprint 4 QA Full Pipeline Test Report (84 test cases). All test cases executed on the live HuggingFace Spaces deployment.

TEST AREA	TOTAL	PASS	FAIL	NOTES
Step 6 — Explainability	30	30	0	Feature importance, waterfall, what-if, clinical sense-check, domain coverage
Step 7 — Ethics & Bias	24	24	0	Subgroup table, bias banner, EU AI Act checklist, representation chart, case studies
Certificate / E2E / Deliverable	30	30	0	PDF generation, end-to-end pipeline, cross-step consistency, domain switching
TOTAL	**84**	**84**	**0**	**100% PASS RATE**

Individual test case evidence: 84 screenshots captured (TC-S4-001 through TC-S4-084). Original tester: Berfin Duru Alkan. Verified on production environment.

8. Sprint 4 Metrics

METRIC	WHAT WE MEASURED	RESULT	TARGET	STATUS
Bias Detection Accuracy	Subgroup gap at 10pp and >10pp	Banner hidden at ≤10pp; visible at >10pp. Confirmed with Cardiology (25.2pp gap → visible) and Neurology (within threshold → hidden)	Correct threshold behaviour	**PASS**
Checklist Toggle	All 8 EU AI Act items	8 items render, 2 pre-checked on load, all toggle correctly. 3/3 backend checklist tests pass	All toggle; 2 pre-checked	**PASS**
Certificate Content	PDF for 3 different domains	Each PDF includes domain name, model type, 6 metrics (Accuracy, Sensitivity, Specificity, Precision, F1, AUC-	Correct content per domain	**PASS**

METRIC	WHAT WE MEASURED	RESULT	TARGET	STATUS
		ROC), confusion matrix, bias findings, checklist state		
Certificate Generation Time	POST /api/generate-certificate timing	0.69 seconds measured via curl	< 10 seconds	**PASS**
End-to-End Flow	Steps 1–7 with fresh CSV	Full 7-step pipeline completed for 3 domains (Cardiology, Neurology, Endocrinology) with default and uploaded CSV. Zero crashes, all step locks enforced	Zero crashes	**PASS**
Clinical Language Audit	Count raw column names in UI	0 raw database column names observed across all tested domains. All labels use clinical display names	0 raw names	**PASS**
Domain Count (Steps 6–7)	Run Steps 6–7 across domains	Verified for 3 domains (Cardiology, Neurology, Endocrinology). All 20 domains visible in specialty selector. Feature importance and subgroup tables update correctly on switch	20/20	**PASS**

9. Next Week Plan (Sprint 5 — Final Submission)

Sprint 5 scope: **Final Presentation** + **Complete Documentation** + **Polish & Hardening**.

STORY ID	TITLE	ASSIGNEE
US-048	Final presentation slides and live demo preparation	Team
US-049	ISO 42001 report finalisation and submission	Efe Çelik / Berat Mert
US-050	End-to-end regression test across all 20 domains	Berfin Duru Alkan
US-051	UI polish — responsive layout, accessibility audit	Burak Aydoğmuş
US-052	Documentation cleanup — README, wiki, inline comments	Batuhan Bayazıt

10. Retrospective Note (Week 8 — Even Week)

KEEP

QA-driven development with test plan written before implementation continued to catch issues early. The 84-test-case QA report was completed before the deadline with zero failures. Clinical language review during development prevented post-hoc renaming.

IMPROVE

Sprint 4 had a heavy documentation load (ISO 42001 report) alongside feature work. Starting documentation earlier in the sprint would reduce last-day pressure. Jira ticket creation for QA stories could be more granular.

TRY

For Sprint 5, run a full 20-domain regression test as a CI gate before any merge to main. Schedule the final presentation rehearsal at least 2 days before the showcase to allow feedback incorporation.